

Krish Puwar

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EDUCATION

The State University of New York at Buffalo

Master of Science in Computer Science - Computer Security and Privacy Specialization

Buffalo, NY

August 2026 – December 2027

The State University of New York at Buffalo

Bachelor of Science in Computer Engineering — Cum Laude, GPA 3.44 — Dean's List

Buffalo, NY

May 2026

EXPERIENCE

Undergraduate Research Engineer Intern

February 2026 – May 2026

Integrated Systems

Victor, NY

- Contributing to **GPROP**, an RF signal propagation modeling tool built in the WN4SS Lab as part of an **SPIR -funded** industry collaboration simulating multipath propagation in IoT-dense urban environments.
- Authored **Bash automation scripts** chaining GPROP's fetch, preprocessing, and model execution into one pipeline.
- Built **containerized Docker environments** standardizing the pipeline across dev and deployment targets.
- Documented system architecture, pipeline design, and experimental results for a forthcoming **research publication**.

Security Engineering Intern

July 2025 – August 2025

Zayed University

Abu Dhabi, UAE

- Worked at the university's on-campus **Security Operations Center (SOC)**, engineering a full-stack SOC training platform (**Flask, PostgreSQL, Docker**) with a normalized schema and REST API contracts for real-time performance analytics.
- Used **MITRE Caldera** for **red-team attack simulation**, chaining ATT&CK-mapped TTPs to emulate real adversary behavior, paired with **Wazuh** monitoring to drive live detection-and-response exercises.
- Built **AI-driven automation** pipelines with the Gemini API to generate scenario content and score exercises dynamically, reducing manual SOC training overhead.
- Deployed and maintained the platform via **Docker** on university infrastructure, ensuring dev/prod environment parity for scalable multi-user training operations.

PROJECTS

AWS SOC SIEM | AWS (EC2, S3, Lambda, SAM), Splunk, MITRE ATT&CK | [GitHub](#)

June 2026 – August 2026

- Built a **cloud-native SIEM** that ingests security logs into S3, parses and scans them through Lambda functions running **5 MITRE ATT&CK-mapped detection rules**, and stores structured alerts in DynamoDB.
- Engineered a **threat-intelligence enrichment service** integrating the VirusTotal and AbuseIPDB APIs to automatically score detected indicators of compromise (IPs, hashes, domains), flag malicious activity, and block offending IPs via EC2.
- Deployed a self-hosted **Splunk Enterprise** instance on **EC2** and streamed alerts into it through the **HTTP Event Collector (HEC)**, replacing a custom dashboard with Splunk dashboards that map detections to MITRE ATT&CK tactics and techniques for analyst triage.
- Defined the full stack as **infrastructure-as-code** via an AWS SAM template with **least-privilege IAM**, enabling reproducible, version-controlled deployments across environments.

SQL Injection Attack & Defense Lab | PHP, MySQL, Docker, cURL | [GitHub](#)

August 2026 - October 2026

- Deployed a containerized **SEED Labs** environment on a Ubuntu VM (ARM64), configuring **Docker Compose** multi-container networking (web server + MySQL) and resolving LVM storage constraints to support the lab environment.
- Exploited an **authentication bypass** vulnerability in a PHP/MySQL login system using crafted SQL injection payloads (' #), retrieving all employee records without valid credentials.
- Escalated privileges via **UPDATE-statement injection**, hijacking WHERE clauses to modify arbitrary users' salary and password fields from a low-privilege session, including injecting a precomputed **SHA1** hash to achieve full account takeover.
- Fixed the **SQL injection** vulnerability on the web server by implementing **prepared statements**, verifying that previously successful injection payloads were neutralized.

RoboFlock – Senior Capstone | ROS2, Python, Bash, Linux | [Documentation](#)

August 2025 – May 2026

- Configured **systemd services** on an NVIDIA Jetson Orin Nano (Linux) to autostart the full ROS2 stack on boot, enabling reliable headless operation without manual intervention.
- Automated the **ROS2 build and launch workflow** with **Bash** scripts, reducing manual setup and deployment errors.
- Developed **Python** driver scripts and tuned **ODrive** motor controllers for stable closed-loop differential-drive control.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, C, Java

Security: SIEM (Wazuh), Threat Intelligence, MITRE ATT&CK, Adversary Emulation (Caldera), Cryptography, OpenSSL

Cloud: AWS (EC2, S3, Lambda, DynamoDB, IAM, SAM), Infrastructure as Code, CloudFormation

Virtualization: Nutanix (Cloud Infrastructure, AHV, Cloud Manager, Kubernetes Platform), Hyper-V, vSphere

Networking: TCP/IP, Network Protocols, iptables, Firewall Configuration, Mininet, Log Analysis, Security Hardening

Data & Databases: PostgreSQL, CodeQL, Pandas, NumPy, Splunk, Tableau, Power BI, Query Optimization, Indexing

Systems & DevOps: Linux, Bash Scripting, Python Scripting, Docker, Git

Certifications: CompTIA Security+ (SY0-701) *Expected 2026* | CCNA *Expected 2026* | AWS CLF 002 *October 2026*